Computational Astrophysics and Geophysics Assignment 3

submitted by

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PLUTO Code

PLUTO is a software for "solving mixed hyperbolic/parabolic systems of partial differential equations (conservation laws) targeting high Mach number flows in astrophysical plasma dynamics". It supports a wide range of numerical schemes, adaptive mesh refinement, and various geometries, making it suitable for studying astrophysical flows and instabilities.

The code uses a finite-volume/finite-difference, shock-capturing approach designed to integrate a system of conservation laws, generally written as:

$$\frac{\partial \boldsymbol{U}}{\partial t} + \nabla \cdot T_h = \nabla \cdot T_p + \boldsymbol{S}(\boldsymbol{U})$$

where $\boldsymbol{U}$ represents a set of conservative quantities, $T_h(\boldsymbol{U})$ is the (hyperbolic) flux tensor, T_p is the diffusion (parabolic) flux tensor and $\boldsymbol{S}(\boldsymbol{U})$ defines the source terms. An equivalent set of primitive variables $\boldsymbol{V}$ is more conveniently used for assigning initial and boundary conditions. The explicit form of $\boldsymbol{U}$, $\boldsymbol{V}$, $T(\boldsymbol{U})$ and $\boldsymbol{S}(\boldsymbol{U})$ depends on the particular physics module selected:

- HD: Newtonian (classical) hydrodynamics
- MHD: Ideal/resistive magnetohydrodynamics
- RHD: Special relativistic hydrodynamics
- RMHD: Special (ideal) relativistic magnetohydrodynamics
- ResRMHD: Special (ideal) relativistic magnetohydrodynamics with resistivity

In this assignment, we have used it to investigate Rayleigh-Taylor Instability using both HD and MHD hydrodynamics. However, since PLUTO is designed for Unix-like systems, running it on Windows requires the use of WSL (Windows Subsystem for Linux), which creates a Linux environment intgrated into Windows. Below, I have described the steps I followed to install WSL before installing PLUTO, setting up Rayleigh-Taylor Instability and plotting the simulation results on my Windows system.

Installing WSL and Preparing the Linux Environment

I first installed WSL via the Powershell terminal using:

```
> wsl --install
```

This command installs WSL along with the default Linux distribution, Ubuntu. The Ubuntu terminal is used to access WSL and is the primary terminal I used for all of the following steps. I first updated and upgraded my system to ensure I have the latest packages:

```
> sudo apt update
> sudo apt upgrade -y
```

According to the PLUTO code website (`http://plutocode.ph.unito.it/`), the following dependencies are required:

- Minimal prerequisites: C compiler, GNU make, Python
- Parallel builds: MPI library
- AMR: Chombo library, C++ compiler, Fortran compiler, HDF5 library

For my purposes, I installed the dependencies needed for basic PLUTO and parallel processing, which include a C compiler (`build-essential`), GNU make (`make`), Python (`python3`), MPI-library (`mpich`) and Fortran complier (`gfortran`):

```
> sudo apt install build-essential make python3 mpich gfortran -y
```

Downloading and Installing PLUTO

I then downloaded the latest version of the PLUTO software (4.4-patch3) from the website via the Ubuntu terminal, so that the file was saved in the same directory.

```
> wget https://plutocode.ph.unito.it/Downloads/pluto-4.4-patch3.tar.gz
```

This downloads an archive file named pluto-4.4-patch3.tar.gz. I extracted this file and entered PLUTO as my working directory using:

```
> tar xvzf pluto-4.4-patch3.tar.gz
> cd PLUTO
```

This folder contains Config (machine architecture and dependency files), Doc (documentation), Lib (repository of libraries, Src (source files), Tools (Python and visualisation tools) and Test Problems (pre-defined setup files for a set of test problems). For the sake of this assignment, we are interested in the Rayleigh-Taylor test problem, located in:

```
> cd ~/PLUTO/Test_Problems/MHD/Rayleigh-Taylor
```

Setting up Rayleigh-Taylor Instability

Theory

The Rayleigh-Taylor (RT) instability problem studies the behaviour of an interface between two fluids of different densities under the influence of gravity, where the denser fluid rests on the rarer one. The Navier-Stokes equation for a compressible, inviscid fluid (i.e. ignoring viscosity for simplicity) is:

$$\rho\frac{D\vec{v}}{Dt} = -\nabla p + \rho g$$

Where ρ is the fluid density, $\vec{v}$ is the fluid velocity, p is pressure, and g is the acceleration due to gravity. At equilibrium, $\vec{v} = 0$, so $D\vec{v}/Dt = 0$. Hence :

$$\nabla p = \rho g \tag{1}$$

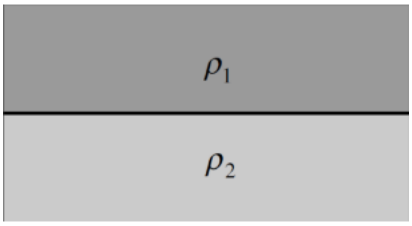

Figure 1: Setup of Rayleigh-Taylor Instability ($\rho_1 > \rho_2$)

Due to symmetry (Fig 1), our problem only has a pressure gradient along the y-axis, so we simply have $dp/dy = \rho g$, which can be integrated to get:

$$p = p_0 + \rho y g \tag{2}$$

The initial density profile (specified in `init.c`) is chosen to be:

$$\rho(y) = \begin{cases} 1 & \text{for } y \leq 0, \\ \eta & \text{for } y > 0 \end{cases} \tag{3}$$

where η is the density of the fluid on top normalized to the value in the rarer fluid. The pressure at the interface p_0 is chosen so that the sound speed in the light fluid is 1.

For magnetized setups (MHD), the magnetic field is purely horizontal $\vec{B} = \chi B_c \hat{i}$, where B_c is the critical magnetic field above which perturbations parallel to the magnetic field are suppressed and χ is a user-specified coefficient:

$$B_c \equiv \sqrt{(\rho_{\text{hi}} - \rho_{\text{lo}})L|g|} = \sqrt{(\eta - 1)|g|} \tag{4}$$

Setup

The Rayleigh-Taylor setup has been tested with the following configurations in the PLUTO test problems set, and hence have pre-defined `definitions.h` and `pluto.ini` files:

Conf.	PHYSICS	GEOMETRY	DIM	T. STEP.	INTERP.	divB	Notes
01	HD	CARTESIAN	2	HANCOCK	LINEAR	-	-
02	HD	CARTESIAN	2	RK3	MP5_FD	-	-
03	HD	CARTESIAN	2	RK3	PARABOLIC	-	-
04	HD	CARTESIAN	2	RK2	LINEAR	-	(*)
05	MHD	CARTESIAN	2	HANCOCK	LINEAR	CT	-
06	MHD	CARTESIAN	2	RK3	MP5_FD	GLM	-
07	MHD	CARTESIAN	2	RK3	PARABOLIC	CT	-
08	MHD	CARTESIAN	2	ChTr	PARABOLIC	GLM	(*)
09	MHD	CARTESIAN	3	RK2	LINEAR	GLM	-
10	MHD	CARTESIAN	3	HANCOCK	LINEAR	GLM	(*)

Table 1: Tested configurations for Rayleigh-Taylor setups. (*) Setups for AMR-Chombo.

Since we are interested in comparing the behaviour of hydrodynamics with and without a magnetic field (i.e. MHD and HD, respectively), I chose to use configuration #01 for HD and #05 for MHD, both of which use Hancock time-stepping and linear interpolation to solve the problem in 2D Cartesian coordinates.

In order to do this, I created separate folders for the HD and MHD simulations within the Rayleigh-Taylor directory and copied the appropriate `definitions.h` and `pluto.ini` files into them, as well as the `init.c` file which contains the equations and initial perturbation.

```
> mkdir RT_HD
> mkdir RT_MHD
> cp definitions_01.h pluto_01.ini init.c RT_HD
> cp definitions_05.h pluto_05.ini init.c RT_MHD
```

Within each directory, I used `python3 $PLUTO_DIR/setup.py` to create a makefile with the definitions and initial conditions chosen, and used a series run by selecting `Linux.gcc.defs`. This makefile could then be compiled and run using:

```
> make
> ./pluto
```

In both of the initialization files (`pluto.ini`), the grid dimensions are $x \in [-0.5, 0.5]$ and $y \in [-1, 1]$ (1:2 aspect ratio), with the fluid interface at $y = 0$. The resolution of the grid is 256×512. The boundary conditions are periodic along the x-axis (to maintain symmetry) and reflective along the y-axis (to maintain the interface and to allow vertical motions develop without leaving the domain).

The periodic boundary condition along the x-axis naturally lends itself to using a sinusoidal initial perturbation to trigger the instability. This perturbation has the form:

$$v_y^{(0)} = -10^{-2}\left(1 + \cos\left(\frac{2\pi x}{L_x}\right)\right)\exp\left(-200y^2\right) \tag{5}$$

Note that the wavelength of this wave is L_x, which means that it is the first frequency mode, corresponding to a single bump across the x-domain. The negative sign indicates a downward displacement, which would cause the fluid on top to move into the rarer fluid.

The parameters of the equations defined in `definitions.h` are:

- `ETA` (η): Density of the upper fluid (normalized against the density of the lower fluid)
- `GRAV` (g): Gravity (must be < 0 since it is downwards)
- `CHI` (χ): Magnetic field strength in units of the critical magnetic field B_c.

For both systems, $\eta = 2$ (i.e. $\rho_1 = 2\rho_2$) and $g = -0.1$ which is negative because gravity acts downwards (pulling ρ_1 into ρ_2). The magnitude of acceleration is small enough to keep the flow mildly compressible and numerically stable, but large enough to produce RT growth on achievable simulation times. Based on these values, we can determine the critical magnetic field:

$$B_c = \sqrt{(\eta - 1)|g|} \approx 0.316$$

The magnetic field coefficient χ differs between HD and MHD, which effectively regulates the strength of a magnetic field along the x-axis. For HD, $\chi = 0$ so there is no magnetic field, but χ is non-zero in MHD. $\chi_c = 1$ is the critical point above which perturbations parallel to the magnetic field are suppressed.

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Results

Rayleigh-Taylor Instability with HD

This hydrodynamic (HD) Rayleigh-Taylor instability simulation involves two 2D fluids with a density ratio of 1:2 ($\eta = 2$) under the influence of gravity ($g = -0.1$) in the absence of a magnetic field ($\chi = 0$). The numerical time-stepping used is Hancock with linear interpolation for a duration of 15s.

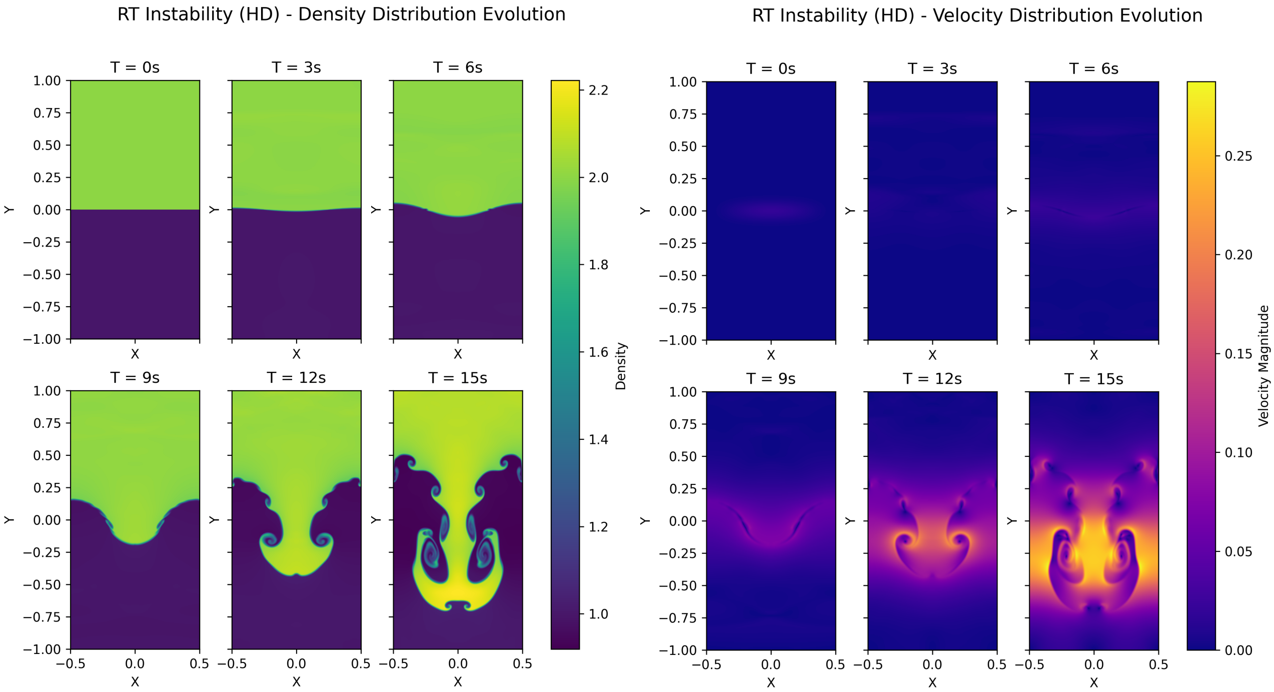

Figure 2: Snapshots of the density $\rho(x, y)$ and velocity magnitude $|\vec{v}(x, y)|$ at 3s intervals for RT instability in the absence of a magnetic field ($\chi = 0$)

Initially, the fluid interface is nearly flat with minor perturbations visible in the velocity distribution. The perturbations begin to grow, forming small undulations as the heavier fluid sinks downward in the middle and the lighter one rises along the sides. Eventually, the density plots show a distinct mushroom-like plume developing as shear forces along the fluid interface cause small scale vortices to form, promoting further mixing. The velocity maps show strong downward flows centred in the plume as the denser fluid is accelerated downwards, and upward flows on either side as the rarer fluid is displaced. Notably, in the absence of any magnetic field, local instabilities are allowed to freely develop into secondary and tertiary spiralling vortices.

Rayleigh-Taylor Instability with MHD

This magneto-hydrodynamic (MHD) Rayleigh-Taylor instability simulation involves two 2D fluids with a density ratio of 1:2 ($\eta = 2$) under the influence of gravity ($g = -0.1$) in the presence of a magnetic field. As we determined earlier, the critical magnetic field $B = B_c$ is achieved when $\chi = 1$. Here, I have investigated the range $\chi = 0.2, 0.5, 0.8$ and 1.2. The numerical time-stepping used is Hancock with linear interpolation for a duration of 15s.

Below, I have plotted the time evolution for $\chi = 0.2$ and $\chi = 1.2$ to illustrate the difference between the subcritical and supercritical behaviour of the RT instability in the presence of a magnetic field.

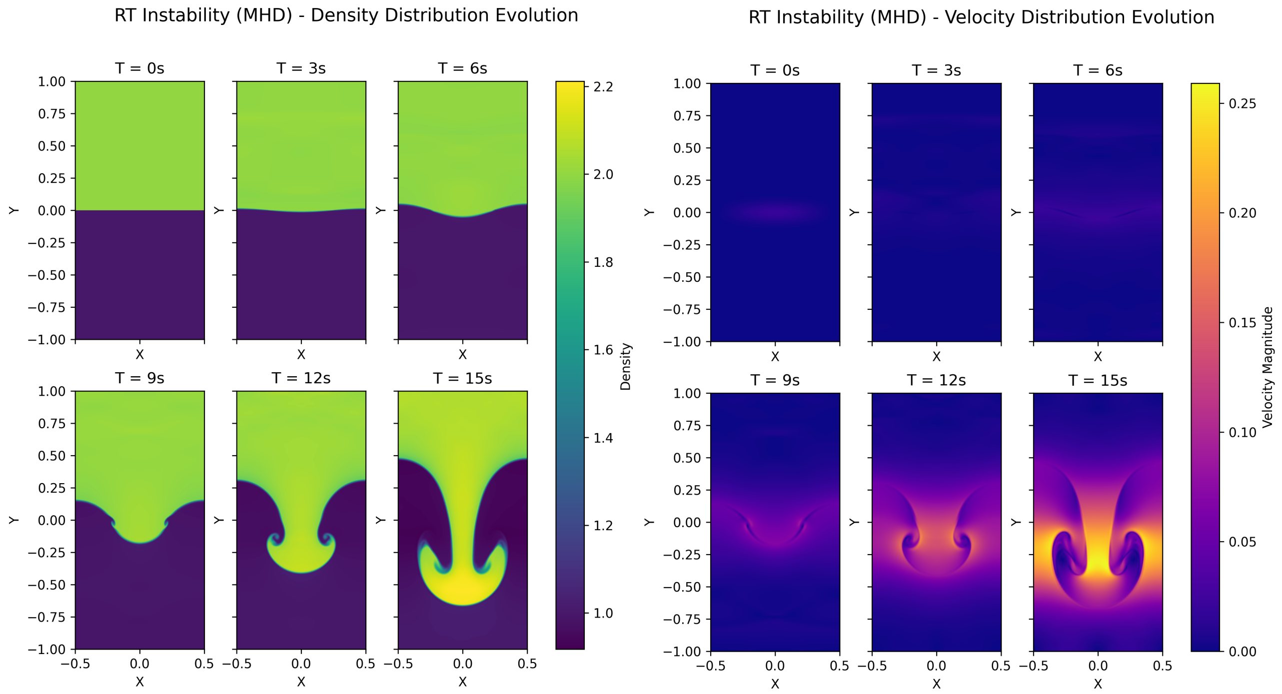

Figure 3: Snapshots of the density $\rho(x, y)$ and velocity magnitude $|\vec{v}(x, y)|$ at 3s intervals for RT instability in the presence of a weak horizontal magnetic field ($\chi = 0.2$)

The initial stages show no deviation from the HD case, with minor velocity perturbations and a nearly flat interface. The growth of the instability is notably slower in the presence of a magnetic field, but the large scale mushroom-like plume still develops. However, the density plots no longer show any small-scale vortices along the interface and therefore there is minimal mixing of the two fluids. The velocity maps also show a more uniform and stable plume formation, with no small-scale perturbations.

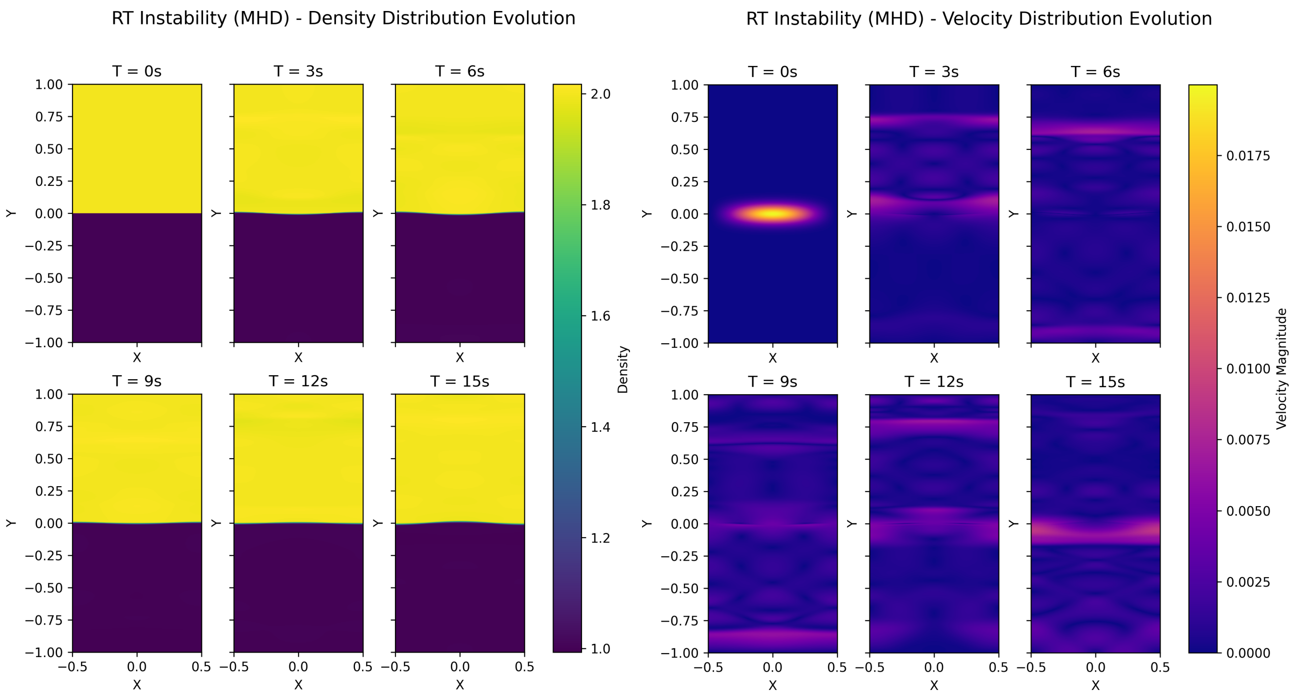

Figure 4: Snapshots of the density $\rho(x, y)$ and velocity magnitude $|\vec{v}(x, y)|$ at 3s intervals for RT instability in the presence of a strong horizontal magnetic field ($\chi = 1.2$)

When $\chi > 1$, the critical magnetic field is exceeded and the magnetic forces become comparable to gravitational buoyancy, effectively damping out any perturbations. As observed in the density plots, this prevents the formation of any large or small scale structures in the fluid, effectively maintaining the interface between the fluids. Meanwhile, the initial perturbation is dissipated over time as we can see in the velocity plot, with scattered and incoherent waves.

We can gain a greater insight into the effect of the magnetic field by visualising the internal magnetic field interaction in the fluid for various external field strengths.

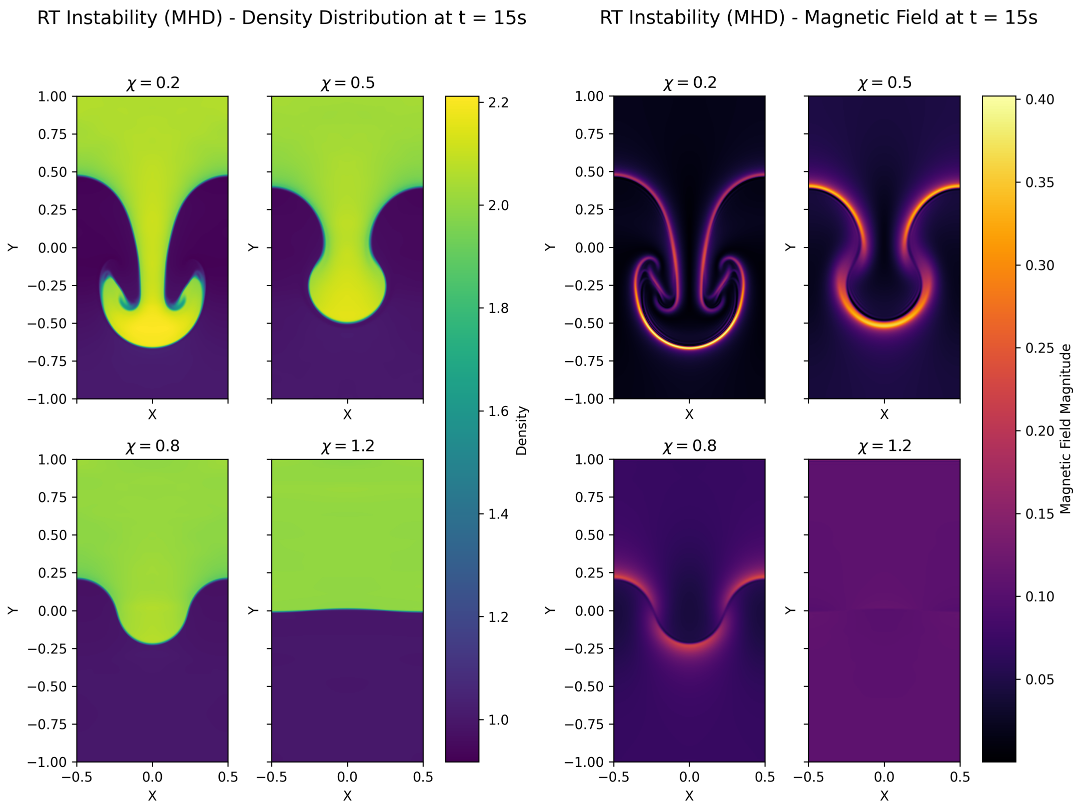

Figure 5: The density $\rho(x, y)$ and magnetic field magnitude $|\vec{B}(x, y)|$ for RT instability with different magnetic field strengths

As we can observe, the magnetic field appears to introduce a restoring "magnetic tension" along field lines which resists bending of the interface, suppressing short-wavelength perturbations. Below the critical point, the instability still grows but with a lower growth rate and smoother structures. Beyond the critical point, the perturbations are quickly damped out by the magnetic forces and equilibrium is maintained.